

Memo WTU158 SEM TEST 2 2009

AFDELING A : MEERVOUDIGE KEUSE VRAE - 15 PUNTE

Gebruik 'n sagte potlood.

Vul jou persoonlike inligting in potlood in op die merkleesvorm se **KANT 2**.

Vul jou studentenommer in van bo na onder en kodeer dit.

Indien jou studentenommer verkeerd gekodeer is of nie ingevul is nie, sal jy nie punte vir afdeling A kry nie.

Beantwoord vrae 1 tot 15 op die **MERKLEESVORM** se **KANT 2**.

Indien kant 1 gebruik word sal dit nie nagesien word nie.

Omkring jou antwoorde eers in hierdie vraestel en merk dan die antwoorde op die vorm as jy seker is van jou keuse.

Jy mag nie verkeerde antwoorde uitvee op die merkleesvorm nie.

SECTION A: MULTIPLE CHOICE QUESTIONS - 15 MARKS

Use a soft pencil.

Fill in your personal information in pencil on the optic reader form on **SIDE 2**.

Fill in your student number from top to bottom and then code it.

If you make a mistake when coding your student number or if the student number is not filled in, you will get no marks for section A.

Answer questions 1 to 15 on the **OPTIC READER FORM** on **SIDE 2**.

If side 1 is used it will not be marked.

First circle your answers in this paper and only mark the answers on the form when you are certain of your choice.

You may not erase wrong answers on the optic reader form.

Vraag 1 / Question 1

Die vergelyking van die raaklyn aan die grafiek van $f(x) = x^3$ by $x = 2$ is

The equation of the tangent line to the graph of $f(x) = x^3$ at $x = 2$ is

[1a] $y - 8 = 3x^2(x - 2)$	[1b] $y - 2 = 3x^2(x - 8)$	☒ [1c] $y - 8 = 12(x - 2)$
[1d] $y - 2 = 12(x - 8)$	[1e] Geen van hierdie / None of these	

Vraag 2 / Question 2

Jy kan 'n raaklyn aan die grafiek van die funksie $y = \ln x$ trek wat ewewydig is aan die lyn $y = 2x + 3$ by die punt waar $x =$

You can draw a tangent line to the graph of the function $y = \ln x$ that is parallel to the line $y = 2x + 3$ at the point where $x =$

☒ [2a] $\frac{1}{2}$	[2b] e^2	[2c] 2	[2d] Geen van hierdie / None of these
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Vraag 3 / Question 3

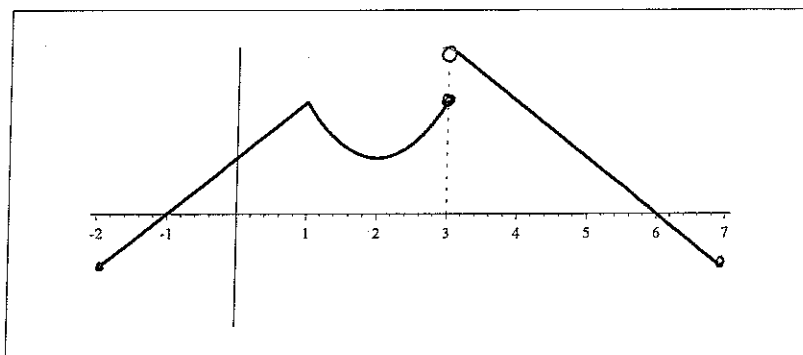
Jy kan 'n raaklyn aan die grafiek van die funksie $y = \sin 2x$, $x \in \left[0, \frac{\pi}{2}\right]$ trek wat horisontaal is by die punt waar $x =$

You can draw a tangent line to the graph of the function $y = \sin 2x$, $x \in \left[0, \frac{\pi}{2}\right]$ that is horizontal at the point where $x =$

[3a] 0	☒ [3b] $\frac{\pi}{4}$	[3c] $\frac{\pi}{2}$	[3d] Geen van hierdie / None of these
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GEBRUIK DIE GRAFIEK VAN DIE FUNKSIE f HIERONDER OM VRAE 4 TOT 7 TE BEANTWOORD.

USE THE GRAPH OF THE FUNCTION f BELOW TO ANSWER QUESTIONS 4 TO 7.



Vraag 4 / Question 4

By watter van die waardes van x , $x \in \{1, 2, 3\}$, is f kontinu? Slegs by

At which of the values of x , $x \in \{1, 2, 3\}$, is f continuous? Only at

[4a] $x = 1$	[4b] $x = 2$	[4c] $x = 3$	☒ [4d] $x = 1, x = 2$
[4e] $x = 1, x = 3$	[4f] $x = 2, x = 3$	[4g] Geen van hierdie / None of these	

Vraag 5 / Question 5

By watter van die waardes van x , $x \in \{1, 2, 3\}$, is f differensieerbaar? Slegs by

At which of the values of x , $x \in \{1, 2, 3\}$, is f differentiable? Only at

[5a] $x = 1$	☒ [5b] $x = 2$	[5c] $x = 3$	[5d] $x = 1, x = 2$
[5e] $x = 1, x = 3$	[5f] $x = 2, x = 3$	[5g] Geen van hierdie / None of these	

Vraag 6 / Question 6

Beskou die bewerings:

I f het 'n absolute / globale minimum by $x = -2$. ✓

II f het 'n absolute / globale maksimum by $x = 3$. ✗

III f het 'n lokale / relatiewe minimum by $x = 2$. ✓

Watter bewering(s) is waar?

Consider the statements:

I f has an absolute / global minimum at $x = -2$. ✓

II f has an absolute / global maximum at $x = 3$. ✗

III f has a local / relative minimum at $x = 2$. ✓

Which statement(s) is(are) true?

[6a] slegs / only I	[6b] slegs / only II	[6c] slegs / only III
[6d] slegs / only I en/and II	☒ [6e] slegs / only I en/and III	[6f] slegs / only II en/and III
[6g] Geen van hierdie / None of these		

Vraag 7 / Question 7

Beskou die bewerings: / Consider the statements:

- I f is kontinu van links in $x = 3$. / f is continuous from the left at $x = 3$. ✓
 II f is kontinu van regs in $x = 3$. / f is continuous from the right at $x = 3$. ✗
 III f is kontinu van links in $x = -2$. / f is continuous from the left at $x = -2$. ✗

Watter bewering(s) is waar? / Which statement(s) is(are) true?

[7a] slegs / only I	[7b] slegs / only II	[7c] slegs / only III
[7d] slegs / only I en/and II	[7e] slegs / only I en/and III	[7f] slegs / only II en/and III
[7g] Geen van hierdie / None of these		

Vraag 8 / Question 8

Die grootste moontlike interval waarop die funksie $f(x) = \frac{x}{9-x^2} + \sqrt{x}$ kontinu is, is

The largest possible interval on which the function $f(x) = \frac{x}{9-x^2} + \sqrt{x}$ is continuous, is

[8a] $(-\infty, \infty)$	[8b] $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$	[8c] $(-3, 3)$
[8d] $[0, \infty)$	[8e] $[0, 3) \cup (3, \infty)$	[8f] Geen van hierdie / None of these

Vraag 9 / Question 9

Beskou die bewerings:

- I Indien f kontinu is in $x = a$, dan is f differensieerbaar in $x = a$.
 II Indien f differensieerbaar is in $x = a$, dan is f kontinu in $x = a$. ✓
 III Indien f nie kontinu is in $x = a$ nie, dan is f nie differensieerbaar in $x = a$ nie. ✓

Watter bewering(s) is waar?

Consider the statements:

- I If f is continuous in $x = a$, then f is differentiable at $x = a$.
 II If f is differentiable in $x = a$, then f is continuous at $x = a$. ✓
 III If f is not continuous in $x = a$, then f is not differentiable in $x = a$. ✓

Which statement(s) is(are) true?

[9a] slegs / only I	[9b] slegs / only II	[9c] slegs / only III
[9d] slegs / only I en/and II	[9e] slegs / only I en/and III	[9f] slegs / only II en/and III
[9g] Geen van hierdie / None of these		

Vraag 10 / Question 10

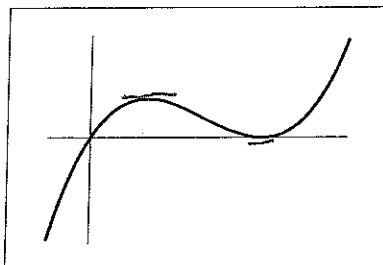
$\sinh(x) - \cosh(x) =$

[10a] e^x	[10b] e^{-x}	[10c] 0	[10d] $\frac{1}{2}e^x$	[10e] $\frac{1}{2}e^{-x}$
[10f] Geen van hierdie / None of these				

Vraag 11 / Question 11

Beskou die grafiek van die funksie f .

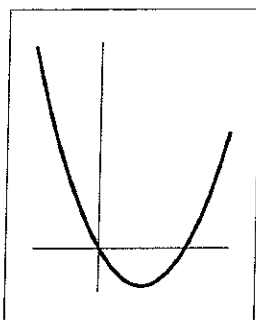
Consider the graph of the function f .



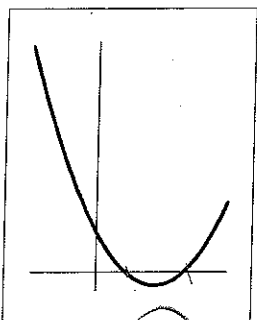
Funksie / Function f

Die grafiek van die afgeleide funksie $f'(x)$ is

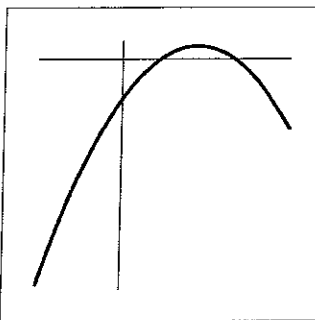
The graph of the derivative function $f'(x)$ is



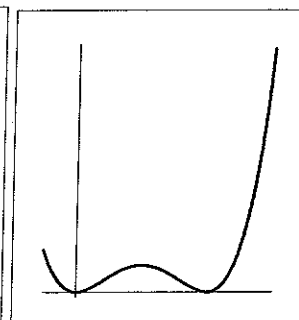
[11a]



[11b]



[11c]



[11d]

[11e] Geen van hierdie / None of these

Vraag 12 / Question 12

Indien jy die definisie gebruik om die afgeleide van $f(x) = e^{2x}$ in $x = 3$ te bepaal, watter van die limiete hieronder is gelyk aan $f'(3)$?

If you use the definition to find the derivative of $f(x) = e^{2x}$ at $x = 3$, which of the limits below are equal to $f'(3)$?

Limiete: / Limits:

I $\lim_{h \rightarrow 0} \frac{e^{6+h} - e^6}{h}$

II $\lim_{h \rightarrow 0} \frac{e^{6+2h} - e^6}{h}$

III $\lim_{x \rightarrow 3} \frac{e^{2x} - e^6}{x - 3}$

[12a] slegs / only I	[12b] slegs / only II	[12c] slegs / only III
[12d] slegs / only I en/and II	[12e] slegs / only I en/and III	[12f] slegs / only II en/and III
[12g] Geen van hierdie / None of these		

Vraag 13 / Question 13

Die funksie $f(x) = \left(\frac{x-2}{2x+1}\right)^2$ het slegs een kritieke punt $x = 2$.

Die globale/absolute maksimum van f op die interval $[-1, 1]$

The function $f(x) = \left(\frac{x-2}{2x+1}\right)^2$ has only one critical point $x = 2$.

The global/absolute maximum of f on the interval $[-1, 1]$

[13a] = 0	[13b] = 9	[13c] = $\frac{1}{9}$
[13d] bestaan nie want die funksie het 'n vertikale asimptoot by $x = -\frac{1}{2}$		
[13d] doesn't exist because the function has a vertical asymptote at $x = -\frac{1}{2}$		
[13e] Geen van hierdie / None of these		

Vraag 14 / Question 14

Indien $f(x) = \frac{x}{2+x}$ dan is $f'(x) = \frac{2}{(2+x)^2}$.

Die funksie is stygend op die interval

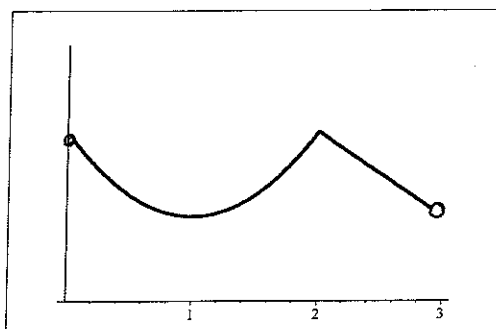
If $f(x) = \frac{x}{2+x}$ then $f'(x) = \frac{2}{(2+x)^2}$.

The function is increasing on the interval

[14a] $(-\infty, \infty)$	[14b] $(-\infty, -2)$	[14c] $(-2, \infty)$	[14d] $(-2, 2)$
[14e] $(-\infty, -2) \cup (-2, \infty)$ [14f] Geen van hierdie / None of these			

Vraag 15 / Question 15

Beskou die grafiek van die funksie f . / Consider the graph of the function f .



Beskou die intervalle: / Consider the intervals: .

I $[0, 2]$ II $[1, 3]$ III $[2, 3]$

Die funksie voldoen aan die voorwaardes van die Middelwaardestelling op die interval(le)

The function satisfies the conditions of the Mean Value Theorem on the interval(s)

[15a] slegs / only I	[15b] slegs / only II	[15c] slegs / only III
[15d] slegs / only I en/and II	[15e] slegs / only I en/and III	[15f] slegs / only II en/and III
[15g] Geen van hierdie / None of these		

[15]

AFDELING B: MAKSIMUM 30 PUNTE

BEANTWOORD ALLE VERDERE VRAE OP HIERDIE VRAESTEL IN INK.

SECTION B: MAXIMUM 30 MARKS

ANSWER ALL THE FOLLOWING QUESTIONS ON THIS PAPER IN INK.

Vraag 16 / Question 16

- i Voltooi die definisie: 'n Funksie f is kontinu in $x = a$ as ...

Complete the definition: A function f is continuous at $x = a$ if ..

$$\lim_{x \rightarrow a} f(x) = f(a)$$

[1]

- ii Vir watter waarde(s) van a en b sal die funksie f kontinu wees in $x = 4$?

Toon duidelik aan hoe u die definisie in Vraag i gebruik.

For which value(s) of a and b will the function f be continuous at $x = 4$?

Clearly show how you use the definition in Question i.

$$f(x) = \begin{cases} ax + (5 - 2a) & \text{as / if } x < 4 \\ 6 & \text{as / if } x = 4 \\ b(x - 6)^2 + 3 & \text{as / if } x > 4 \end{cases}$$

As $\lim_{x \rightarrow 4} f(x)$ exists iff $\lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^-} f(x)$ it

follows from i that

$$\bullet \lim_{x \rightarrow 4^+} f(x) = f(4) \Rightarrow \lim_{x \rightarrow 4^+} b(x - 6)^2 + 3 = 6$$

$$\Rightarrow 4b + 3 = 6 \Rightarrow 4b = 3 \Rightarrow b = \frac{3}{4}$$

AND

$$\bullet \lim_{x \rightarrow 4^-} f(x) = f(4) \Rightarrow \lim_{x \rightarrow 4^-} ax + (5 - 2a) = 6$$

$$\Rightarrow 4a + (5 - 2a) = 6 \Rightarrow 2a = 1 \Rightarrow a = \frac{1}{2}$$

[3]

Vraag 17 / Question 17

Bepaal die volgende afgeleides. Moenie jou antwoorde vereenvoudig nie.

Find the following derivatives. Do not simplify your answers.

i $f'(t)$ as / if $f(t) = 2^{3t} + 3e^{-2t}$.

$$f'(t) = (\ln 2) \times 3 \times 2^{3t} + 3 \cdot (-2) e^{-2t}$$

[2]

ii $f'(x)$ as / if $f(x) = 3 \cot x \sin 5x$.

$$f'(x) = 3(-\operatorname{cosec}^2 x) \sin 5x + 3 \cot x \times 5 \cos 5x$$

[2]

iii $\frac{ds}{dz}$ as / if $s = \sin^{-1}(1 - 2z)$.

(Notasie: $\sin^{-1} = \operatorname{bsin}$ / Notation: $\sin^{-1} = \operatorname{arcsin}$)

$$\frac{ds}{dz} = \frac{1}{\sqrt{1 - (1 - 2z)^2}} \quad (-2)$$

[1]

iv $\frac{dy}{dx}$ as / if $x^4 + xy^3 = \tan y$.

$$\frac{d}{dx}(x^4 + xy^3) = \frac{d}{dx} \tan y$$

$$\Rightarrow 4x^3 + 1 \cdot y^3 + x \cdot 3y^2 \cdot \frac{dy}{dx} = \sec^2 y \cdot \frac{dy}{dx}$$

$$\Rightarrow 3xy^2 \frac{dy}{dx} - \sec^2 y \frac{dy}{dx} = -4x^3 - y^3$$

[3]

$$\Rightarrow \frac{dy}{dx} = \frac{-4x^3 - y^3}{3xy^2 - \sec^2 y}$$

v $\frac{dy}{dx}$ as / if $y = (\cos x)^{\frac{1}{x}}$.

$$\ln y = \ln(\cos x)^{\frac{1}{x}} = \frac{1}{x} \ln \cos x = \frac{\ln \cos x}{x}$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{\frac{-\sin x}{\cos x} \cdot x - \ln \cos x \cdot 1}{x^2}$$

$$\therefore \frac{dy}{dx} = (\cos x)^{\frac{1}{x}} \left[\frac{-\sin x}{\cos x} \cdot x - \ln \cos x \right] \cdot \frac{1}{x^2}$$

OR

$$\ln y = \frac{1}{x} \ln \cos x$$

$$\frac{1}{y} \frac{dy}{dx} = \left(-\frac{1}{x^2}\right) \ln \cos x + \frac{1}{x} \frac{-\sin x}{\cos x}$$

$$\therefore \frac{dy}{dx} = (\cos x)^{\frac{1}{x}} \left[-\frac{1}{x^2} \ln \cos x + \frac{1}{x} \frac{-\sin x}{\cos x} \right]$$

[3]

Vraag 18 / Question 18

Bepaal, indien dit bestaan, al die horisontale asymptote van $f(x) = \frac{x^2 - 4x^5}{16x^5 + 9x}$.

Find, if it exists, all the horizontal asymptotes of $f(x) = \frac{x^2 - 4x^5}{16x^5 + 9x}$.

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{\frac{1}{x^3} - 4}{16 - \frac{9}{x^4}} = -\frac{4}{16} = -\frac{1}{4}$$

and

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} \frac{\frac{1}{x^3} - 4}{16 - \frac{9}{x^4}} = -\frac{4}{16}$$

$y = -\frac{1}{4}$ is the only horizontal asymptote

[2]

Vraag 19 / Question 19

Bepaal die kritieke getalle van $f(x) = x - \sqrt{1+4x}$.

Find the critical numbers of $f(x) = x - \sqrt{1+4x}$.

$$\begin{aligned} f'(x) &= 1 - \frac{1}{2}(1+4x)^{-\frac{1}{2}}(4) \\ &= 1 - \frac{2}{\sqrt{1+4x}} \end{aligned}$$

$$f'(x)=0 \Rightarrow 1 = \frac{2}{\sqrt{1+4x}}$$

$$\Rightarrow \sqrt{1+4x} = 2$$

$$\Rightarrow 1+4x = 4$$

$$\Rightarrow 4x = 3$$

$$\Rightarrow x = 3/4$$

(f' does not exist for $x = -\frac{1}{4}$, but as the domain is $[-\frac{1}{4}, \infty)$ f cannot have a local extreme)

[3]

Vraag 20 / Question 20

at $x = -1/4$

Gebruik implisiete differensiasie om aan te toon dat

$$\frac{d}{dx} \sin^{-1}(2x) = \frac{d}{dx} \arcsin(2x) = \frac{2}{\sqrt{1-4x^2}}.$$

Use implicit differentiation to show that $\frac{d}{dx} \sin^{-1}(2x) = \frac{d}{dx} \arcsin(2x) = \frac{2}{\sqrt{1-4x^2}}.$

$$y = \sin^{-1} 2x \Rightarrow \sin y = 2x \quad \text{and} \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\Rightarrow \cos y \times \frac{dy}{dx} = 2$$

$$\Rightarrow \frac{dy}{dx} = \frac{2}{\cos y}$$

$$= \frac{2}{\sqrt{1-\sin^2 y}}$$

because $\cos y$ is pos on $[-\frac{\pi}{2}, \frac{\pi}{2}]$

$$= \frac{2}{\sqrt{1-(2x)^2}} = \frac{2}{\sqrt{1-4x^2}}.$$

[3]

Vraag 21 / Question 21

Beskou die funksies f, f' en f'' .

Consider the functions f, f' and f'' .

$$f(x) = 2x^{\frac{5}{3}} - 5x^{\frac{4}{3}} \quad f'(x) = \frac{10x^{\frac{1}{3}}(x^{\frac{1}{3}} - 2)}{3} \quad f''(x) = \frac{20(x^{\frac{1}{3}} - 1)}{9x^{\frac{2}{3}}}$$

- i Die funksie het 'n kritieke getal $x = 0$.

Gebruik die Eerste Afgeleide Toets om te besluit of die funksie 'n lokale/relatiewe minimum of maksimum by $x = 0$ het.

Toon DUIDELIK aan hoe jy die toets gebruik.

The function has a critical number $x = 0$.

Use the First Derivative Test to decide whether the function has a local / relative minimum or maximum at $x = 0$.

CLEARLY show how you use the test.

The critical numbers are 0 and 8

• For $c = -1 < 0$ $f'(-1) = \frac{10 \times (-1) \times (-3)}{3} > 0$

∴ f is increasing on $(-\infty, 0)$

• For $c = 1 \in (0, 8)$, $f'(1) = \frac{10 \times 1 \times (-1)}{3} < 0$

∴ f is decreasing on $(0, \infty)$

As f' changes from positive to negative at $x = 0$, f has a local maximum at $x = 0$.

[3]

- ii Die funksie het 'n ook 'n kritieke getal $x = 8$.

Gebruik die Tweede Afgeleide Toets om te besluit of die funksie 'n lokale/relatiewe minimum of maksimum by $x = 8$ het.

Toon DUIDELIK aan hoe jy die toets gebruik.

The function also has a critical number $x = 8$.

Use the Second Derivative Test to decide whether the function has a local / relative minimum or maximum at $x = 8$.

CLEARLY show how you use the test.

$$f''(8) = \frac{20 \times (8^{\frac{1}{3}} - 1)}{9 \times 8^{\frac{2}{3}}} = \frac{20 \times 1}{9 \times 4} > 0$$

∴ f is concave up on a small interval around $x = 8$

[2]

∴ f has a local minimum at $x = 8$.

Vraag 22 / Question 22

Bepaal die volgende limiete, indien dit bestaan. / Calculate the following limits, if it exists.

i $\lim_{x \rightarrow 2} \frac{\sin(2-x)}{e^{x-2} - 1}$ (form $\frac{0}{0}$)

$= \lim_{x \rightarrow 2} \frac{\cos(2-x)(-1)}{e^{x-2}}$ (using L' Hospital)

$= \frac{\cos 0 \cdot -1}{e^0}$

$= -1$

[2]

ii $\lim_{x \rightarrow -\infty} (e^{-2x} + 3)^{\frac{1}{x}}$ (form ∞^0)

let $y = \lim_{x \rightarrow -\infty} (e^{-2x} + 3)^{\frac{1}{x}}$

$\ln y = \lim_{x \rightarrow -\infty} \frac{1}{x} \ln(e^{-2x} + 3)$ (form $0 \cdot \infty$)

$= \lim_{x \rightarrow -\infty} \frac{\ln(e^{-2x} + 3)}{x}$ (form $\frac{\infty}{-\infty}$)

$= \lim_{x \rightarrow -\infty} \frac{\frac{-2e^{-2x}}{e^{-2x} + 3}}{1}$

using L' Hospital!

$= \lim_{x \rightarrow -\infty} \frac{-2}{1 + 3/e^{2x}} = -2$

[3]

$y = e^{\ln y} = e^{-2}$

Memo Short questions

Question 1

$$\left. \begin{array}{l} f(x) = x^3 \quad \therefore f(2) = 2^3 = 8 \\ f'(x) = 3x^2 \quad \therefore f'(2) = 3 \times 4 = 12 \end{array} \right\} \Rightarrow \text{equation is } y - 8 = 12(x - 2)$$

Question 2

$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$$

$$f'(x) = 2 \Rightarrow \frac{1}{x} = 2 \Rightarrow x = \frac{1}{2}$$

Question 3

$$f(x) = \sin 2x \Rightarrow f'(x) = 2 \cos 2x$$

$$f'(x) = 0 \Rightarrow 2 \cos 2x = 0 \Rightarrow \cos 2x = 0 \Rightarrow 2x = \frac{\pi}{2} \Rightarrow x = \frac{\pi}{4}$$

Question 4

$$\text{at } x = 1, 2$$

Question 5

$$x = 2$$

Question 6

I and III

Question 7

I

Question 8

$\frac{x}{9-x^2}$ is continuous on $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$

$\sqrt{x}$ is continuous on $[0, \infty)$

$\therefore f$ is continuous on $[0, 3) \cup (3, \infty)$

Question 9

II and III

Question 10

$$\begin{aligned}\sinh x - \cosh x &= \frac{1}{2}(e^x - e^{-x}) - \frac{1}{2}(e^x + e^{-x}) \\ &= \frac{1}{2}(e^x - e^{-x} - e^x - e^{-x}) = \frac{1}{2}(-2e^{-x}) = -e^{-x}\end{aligned}$$

Question 11

11(b)

Question 12

$$\begin{aligned}f'(3) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{e^{2(3+h)} - e^{2 \cdot 3}}{h} \\ &= \lim_{h \rightarrow 0} \frac{e^{6+2h} - e^6}{h}\end{aligned}$$

$$\text{or } f'(3) = \lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} = \lim_{x \rightarrow 3} \frac{e^{2x} - e^6}{x - 3}$$

Question 13

$$\lim_{x \rightarrow -\frac{1}{2}} f(x) = \lim_{x \rightarrow -\frac{1}{2}} \left(\frac{x-2}{2x+1} \right) = +\infty \text{ no global max}$$

Question 14

$f'(x) > 0$ for $x \neq -2$, increasing on $(-\infty, -2) \cup (-2, \infty)$

Question 15

f must be cont on $[a, b]$ and diff b on (a, b)